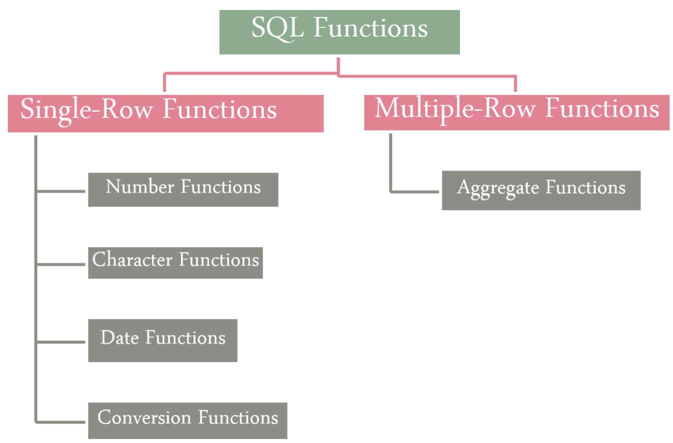
	MARATHWADA INSTITUTE OF TECHNOLOGY, AURANGABAD	**LABORATORY MANUAL**
	PRACTICAL EXPERIMENT INSTRUCTION SHEET	
	EXPERIMENT TITLE**:** Implementation of different types of SQL functions with suitable example 1)Number Function 2) Character Function 3) Aggregate Function 4) Conversion Function 5) Date Function	

DEPARTMENT: COMPUTER SCIENCE AND ENGINEERING		LABORATORY: MIT-CSE-DBIT
EXPERIMENT NO.: 03		YEAR: 2017-18
Class: **T.E.**	PART: I	SUBJECT: **Database Management System**

AIM: Implementation of different types of SQL functions with suitable example
1) Number Function 2) Character Function 3) Aggregate Function
4) Conversion Function 5) Date Function

THEORY:



Single Row Functions: These functions operate on data row-by-row. So, output of these functions will contain number of rows present in input database table.

- **Number Functions:**
  - **ROUND** : ROUND (column_name/constant value, n)

    It round offs the value till 'n' decimal digits
  - **TRUNC** : TRUNC (column_name/constant value, n)

    It truncates the value till 'n' decimal digits
  - **MOD** : MOD(column_name1/value1, column_name2/value2)

Shilpa Sanap			Dr. Smita Kasar
Subject Teacher	**Course Co-ordinator**	**Module Co-ordinator**	**HOD CSE**

Gives remainder value which is calculated after doing column_name1/value1 ÷ column_name2/value2

- **Character Functions:**

– **LOWER :** LOWER(column_name/constant string)

Displays all characters in lower case

– **UPPER :** UPPER(column_name/constant string)

Displays all characters in upper case

– **INITCAP:** INITCAP(column_name/constant string)

Displays first character in upper case

– **LENGTH** : LENGTH(column_name/constant string)

Displays length of string/text

– **CONCAT:** CONCAT(column_name1/constant string1, column_name2/constant string2)

Concatenates first value with second value

– **SUBSTR:** SUBSTR(column_name/constant string, m, n)

Returns characters starting at position m & length n from given value

– **INSTR:** INSTR(column_name/constant string, ‘string’, m , n)

Returns the numeric position of a named ‘string’. Optionally it can start searching from m and occurrence n of a string

- **LPAD/RPAD :** LPAD/RPAD(column_name/constant string, n, 'string')

  Appends the 'string' value to right/left to make maximum length = n

- **TRIM :** TRIM(trim_character from trim_source)

  Trim character from a string

- **REPLACE :** REPLACE(text/column_name, search_string, replacement_string)

  Searches a text expression for a character string and, if found, replaces it with a specified replacement string

■ **Date Functions:**

- **MONTHS_BETWEEN:**MONTHS_BETWEEN(date1/column_name1,date2/column_name2)

  Number of months between two dates

- **ADD_MONTHS:** ADD_MONTHS(date/column_name, n)

  Add number of n calendar months to date

- **NEXT_DAY:** NEXT_DAY(date/column_name, 'day of a week')

  Finds the date of the next specified day of the week

- **LAST_DAY:** LAST_DAY(date/column_name)

  Finds the date of the last day of the month that contains given date

- **ROUND :** ROUND(date, 'format')

  Returns date rounded to the unit specified by the format model

Assume SYSDATE = ‘25-JUL-95’

E.g. ROUND(SYSDATE , ‘Month’) **o/p** : 01-AUG-95

ROUND(SYSDATE, ’Year’) **o/p** : *01-JAN-96*

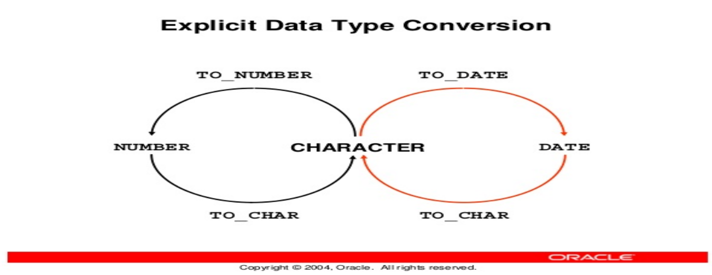
– **TRUNC :** TRUNC(date, ‘format’)

Returns date truncated to the unit specified by the format model

E.g. TRUNC(SYSDATE , ‘Month’) o/p : 01-JUL-95

TRUNC(SYSDATE, ’Year’) o/p : 01-JAN-95

- **Conversion Functions:**



– **TO_CHAR :** TO_CHAR(number/date, ‘format’)

Specified number/date will be converted to given format

– **TO_NUMBER :** TO_NUMBER(character, ‘format’)

Converts a character string containing digits to a number in the format specified by the ‘format’

– **TO_DATE :** TO_DATE(character, ‘format’)

Converts a character string representing a date to a date value according to the format specified-

- **Aggregate/Group/Multiple-Row Functions:**

– **SUM :** SUM(column_name)

Calculate total of values stored in specified column_name

– **COUNT :** COUNT(column_name/*)

Count number of rows which are fetched (those rows are considered where specified column_name value is not null)

– **MAX :** MAX(column_name)

Find maximum value from sepecified column_name

– **MIN :** MIN(column_name)

Find minimum value from sepecified column_name

– **AVG :** AVG(column_name)

Find average value from sepecified column_name

(EXPECTED OBSERVATION / CALCULATION / RESULT)

Sample Output:

Query 1

Calculate the remainder of a salary after it is divided by 500 for all male employees. (MOD)

MOD(SALARY,700)
500
100

Query 2

Display first name of an employee by concatenating it with ‘Mr.’ for all male employees. (CONCAT)

CONCAT('MR.',FNAME)
Mr.Mahesh
Mr.Sandeep

Query 3

Calculate and display length of address and address of an employee. (LENGTH)

ADDRESS	LENGTH(ADDRESS)
Pune	4
Ahmednagar	10
Aurangabad	10
Aurangabad	10

Query 4

Trim and display a word ‘HelloWorld’ by removing first character ‘H’.

TRIM('H'FROM'HELLOWORLD')
elloWorld

Query 5

Find out the date of birth of the oldest employee. (MIN)

MIN(DOB)
25-FEB-78

Query 6

Display first name of an employee whose first name will be last if we arrange it in alphabetic order. (MAX)

MAX(FNAME)
Sushma

Query 7

Display cumulative salary of all employees. (SUM)

SUM(SALARY)
115000

Query 8

Display date of birth of all employees in ‘fmDD Month YYYY’ format TO_CHAR(dob, ‘fmDD Month YYYY’)

TO_CHAR(DOB,'FMDDMONTHYYYY')
10 January 1988
2 May 1987
25 February 1978
4 April 1990

Query 9

Display date of every employee when he/she was 240 months/20 years old. ADD_MONTHS(dob,240)

ADD_MONTHS(DOB,240)
10-JAN-08
02-MAY-07
25-FEB-98
04-APR-10